

AI Evals Cheat Sheet



THREE LEVELS OF AI EVALS

- 3 A/B testing**
cost ★★★★★ frequency ★★☆☆☆
- 2 Model & Human Eval**
cost ★★★★★ frequency ★★☆☆☆
- 1 Unit Tests**
cost ★☆☆☆☆ frequency ★★★★★

START HERE

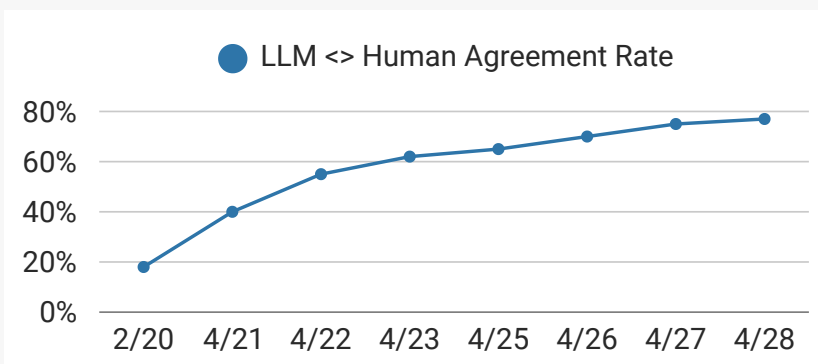


S1: Log Traces
S2: Look at Traces
S3: Model & Human ↻

Step 1: Write Unit Tests
Step 2: Create Test Cases
Step 3: Run & Track Tests Regularly ↻

Write Unit Tests	Test Cases	Run & Track	Log Traces	Look at Traces	Model & Human
- Unlike typical unit tests, organize these assertions for use in other places - For example, data cleaning and automatic retries during inference	- Make them as challenging as possible - Represent real users' interactions - You should not aim for a 100% pass rate. 100% pass is a red flag	- Orchestrate tests in a way that involves the least friction in your tech stack - Start simple and leverage your existing analytics system	- Log sequences of events such as user session or a request flow - Often those are conversations with an LLM - Example: prompt, function, response	- Remove all friction from looking at data. - Render your traces in domain-specific ways - Keep it simple - use what you have before investing	- Automatic evals - LLM as a judge - require humans. - Use binary decisions for clarity - Enhance them with detailed critiques

TRACK MODEL VS. HUMAN EVAL



LET METRICS EMERGE BOTTOM-UP

▼ Top-Down Analysis

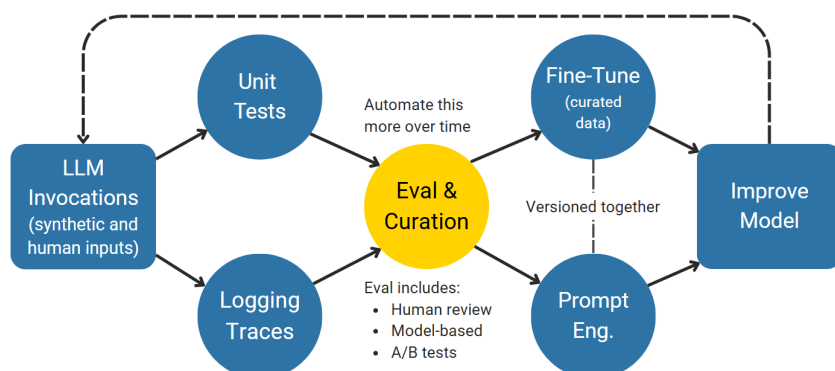
- Starts with metrics like **hallucination** or **toxicity**
- Convenient, but often **misses** domain-specific issues

▲ Bottom-Up Analysis

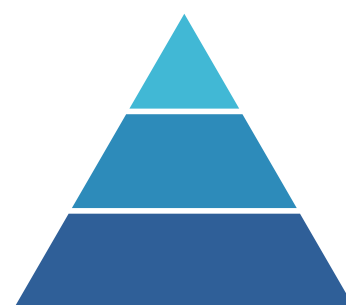
- More **effective**
- Forces you to **look at data**
- Analyze **errors**
- Let **metrics emerge**
- Never stop looking at data

Conclusion: Don't rely on generic evaluation frameworks

AI FLYWHEEL: VIRTUOUS CYCLE



3 SUPERPOWERS EVALS UNLOCK



- Fine-Tuning
- Data Synthesis
- Debugging

Advice: Reuse your existing AI Evals infrastructure